Mustafa Mehboob

Product Engineer · TypeScript · AWS Cloud · Application Security

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Summary

Full-stack product engineer with **3+ years** shipping production SaaS in TypeScript, Node.js/NestJS, React/Next.js and PostgreSQL on AWS. Took over a UK visa platform in the middle of a cyber attack, contained it, rebuilt its AWS architecture, then **led a 3-engineer team that took it past $1M in revenue**. Works with AI coding agents daily and is **first author of a 2026 paper on LLM-based software testing and vulnerability detection**.

Experience

Software Engineer (Team Lead) Jan 2026 – Present
Softaims *Lahore, Pakistan*

- **Took over UKVisaPortal mid-attack**, after a data breach: contained the incident, traced the entry route and closed that whole class of weakness, and kept the service up through a live Layer-7 DDoS, **∼600M of 700M requests blocked at peak**.
- Rebuilt its **AWS architecture**: private VPC with CloudFront VPC Origins → private ALB → ECS Fargate, RDS, WAF and Cloudflare Turnstile, with **AES-256-GCM**-encrypted documents in S3 behind CloudFront signed URLs.
- **Led a 3-engineer team** on the product after recovery: client portal and admin operations suite (auto-assignment and call queues, shift tracking, GDPR/DSAR, Zendesk and Slack); Adyen, MamoPay and PayPal payments; JWT with 4-role RBAC and TOTP MFA. The platform passed **$1M in revenue**.
- **Co-led WorldVisaHub** (Rails 8, Inertia, React 19, PostgreSQL): a versioned drag-and-drop form engine with conditional logic, versions pinned to in-flight orders and in-form payment gating, plus per-user RBAC overrides and a full admin audit trail.
- Built its document pipeline with **zero static cloud keys** (presigned S3 on STS assumed roles, magic-byte validation, KMS, PII-redacted logs) and MediaPipe on-device camera checks; provisioned staging in **Terraform** with an OIDC deploy role, documented in an ADR and runbook.
- Building an industrial **SCADA** backend: NestJS **OPC UA** service (chunked subscriptions, reseed after reconnect, per-node write retry) verified against an 800-tag PLC simulator; Redis pub/sub, TimescaleDB, React 19 HMI; code sample on GitHub.

Associate Software Engineer Feb 2022 – Dec 2025
OptimaGeeks (staff-augmented to Webmended, UAE, Feb 2022 – Feb 2024) *Lahore, Pakistan*

- **Led Userbird**, a GA- and PostHog-style analytics and session-replay platform: injectable JS SDK streaming pageviews, events and recordings into BigQuery; React/Vite dashboard **40% faster** after caching and lazy loading.
- **Led mDrive Asia** e-commerce end to end: Next.js storefront with NextAuth, Django/PostgreSQL backend and admin, Apple GSX and custom payment APIs, from checkout to fulfilment.
- Shipped **8+ production apps**, including a LinkedIn Sales Navigator → CRM Chrome extension (React/TS), WordPress publishing automation (PERN, cron), OpenAI-powered CRM tooling, a bot on Cloudflare Workers, a NestJS backend and a Django QMS, deployed with Docker, PM2 and Nginx.

Selected Project

Mirai – Work Management Suite | *React, TypeScript, Express, Prisma, PostgreSQL, Socket.io, Redis* 2025

- ClickUp-style boards, docs, real-time chat, voice notes and AI assistance across a workspace, team and org hierarchy; board-level RBAC cached in Redis, with live sync over Socket.io.

Publications

First author. "AI-Assisted Automated Software Testing and Vulnerability Detection Using Large Language Models." *Spectrum of Engineering Sciences* 4(3), 2026.
Co-author. "Transfer Learning Optimization for Low-Resource Natural Language Processing Using Transformer Architectures." *Research Consortium Archive* 4(2), 2026.

Technical Skills

Languages: TypeScript, JavaScript, Python, Ruby, SQL **Data**: PostgreSQL, MongoDB, Redis, BigQuery, TimescaleDB
Backend: Node.js, NestJS, Express, Rails, Django, Prisma, Drizzle, REST, webhooks, Socket.io, SSE
Frontend: React, Next.js, Redux, Zustand, Tailwind **AI**: Claude Code (daily), OpenAI API, LLM-assisted testing
Cloud & DevOps: AWS (VPC, ECS Fargate, EC2, RDS, S3, CloudFront, ALB, Route 53, WAF, Shield, KMS, STS, Secrets Manager, GuardDuty, CloudTrail), Terraform, Docker, GitHub Actions, Cloudflare, Nginx
Security & Auth: OAuth 2.0 (NextAuth), Microsoft (Outlook) SSO, JWT, RBAC, TOTP MFA, WAF and DDoS mitigation, KMS and AES-256-GCM encryption, PII redaction, GDPR/DSAR, incident response

Education

Government College University (GCU) Lahore Nov 2021 – May 2025
Bachelor of Science in Computer Science (BSCS), CGPA 3.18 *Lahore, Pakistan*